

Day 12 — Unit 2 : Macro-Economic Analysis

Round 2 · Revision + Practice | UPESSC Asst. Prof. (Economics), Code-43 | Date: 01-08-2026

1. Determination of Output & Employment

Aadhar	Classical	Keynesian	Post-Keynesian
Core Law	Say's Law — supply creates its own demand	Effective Demand — $ADP = ASP$	Income distribution aur growth central
Employment	Hamesha Full Employment; unemployment sirf voluntary/frictional	Underemployment equilibrium possible	Class-based saving; Kaldor, Robinson, Kalecki, Pasinetti
Wage-Price	Poori tarah flexible	Wage downward rigid	Mark-up pricing (Kalecki)
$S = I$ kaise	Via Rate of Interest	Via Level of Income	Via profit share / distribution
Money	Neutral, exogenous	Non-neutral	Endogenous (demand-determined)
Policy	Laissez-faire	Active fiscal policy	Incomes + structural policy

Paradox of Thrift: Keynes mein zyada bachat karne ki koshish → income girta → actual saving utni hi ya kam. Classical mein ye problem nahi (Say's Law + flexible r).

2. Keynes ka Psychological Law of Consumption

- Prop-1: Income ↑ → Consumption ↑, par kam proportion mein ($\Delta C < \Delta Y$)
- Prop-2: $0 < MPC < 1$
- Prop-3: $MPC < APC$, aur income badhne par APC girta jata hai

Multiplier ka aadhar = Prop-2. $K = 1 / (1 - MPC) = 1 / MPS$ | $MPC = 0 \rightarrow K = 1$ · $MPC = 1 \rightarrow K = \infty$

3. Consumption Hypotheses — Char Yaad Rakho

Hypothesis	Economist	Core Idea
Absolute Income	Keynes	$C = f(\text{current absolute } Y)$; APC girta hai jaise Y badhta hai
Relative Income	Duesenberry	Demonstration Effect (padosi ka consumption) + Ratchet Effect (income girne par C downward-rigid, kyunki peak past income ka standard maintain karte hai)
Permanent Income	Friedman	$Y = Y_p + Y_t$; C sirf Y_p se. Transitory income se $MPC \approx 0$ (kyunki Y_t aur C_t mein correlation zero maana)
Life Cycle	Ando – Modigliani – Brumberg	$C = f(\text{lifetime expected resources} = \text{PV of lifetime income} + \text{wealth})$. Young borrow/dissave → middle-age save → retirement dissave

4. Phillips Curve Analysis

- **Original (Phillips, 1958)** — UK data (1861–1957) par **money WAGE rate ke % change** aur unemployment rate ke beech **inverse** sambandh. **Price inflation nahi!**
- **Samuelson & Solow (1960)** — isko **inflation vs unemployment** mein convert kiya → **policy trade-off** menu banaya
- **Friedman – Phelps** — **Expectations-Augmented PC**: $\pi = \pi_e - \beta(U - U_n)$
- **SRPC downward sloping** (money illusion ki wajah se) | **LRPC VERTICAL at NRU**
- **Kyun vertical?** Long-run mein workers expectations adjust kar lete hai → real wage samajh jate hai → higher nominal wage demand → unemployment wapas **Natural Rate** par
- **Accelerationist Hypothesis**: U ko NRU se neeche rakhne ki koshish → sirf **accelerating inflation**, permanent gain nahi
- **Stagflation (1970s)** ne simple Phillips Curve ko demolish kiya

Policy Implication: Unemployment permanently kam karne ke liye sirf **structural / supply-side** measures kaam karenge — demand expansion nahi.

5. Business Cycle Models — Char Model

Model	Core Mechanism	Distinguishing Feature
Samuelson (1939)	Multiplier × Accelerator interaction	Unconstrained cycle. $\alpha = \text{MPC}$, $\beta = \text{accelerator}$ → 5 types of paths. $\alpha\beta < 1 = \text{damped}$ · $\alpha\beta = 1 = \text{constant amplitude}$ · $\alpha\beta > 1 = \text{explosive}$
Hicks (1950)	Multiplier × Accelerator + buffers	Ceiling = full employment output; Floor = autonomous investment + depreciation-limited max disinvestment. Explosive tendency bounded & self-reversing ban jati hai
Kaldor (1940)	Non-linear S aur I functions	S-shaped (non-linear) I & S curves → multiple equilibria (2 stable + 1 unstable). Capital stock key shifting variable
Hawtrey	Bank credit expansion & contraction	Purely Monetary Theory — "cycle is a monetary phenomenon and nothing else". Traders/merchants aur bank credit central

Extra: Schumpeter → **Innovation Theory** of trade cycle (entrepreneur + creative destruction).

6. IS – LM Model (Hicks – Hansen)

- **IS Curve** = **Goods market** equilibrium ($S = I$). **Downward sloping**: $r \downarrow \rightarrow I \uparrow \rightarrow Y \uparrow$
- **LM Curve** = **Money market** equilibrium ($M_d = M_s$). **Upward sloping**: $Y \uparrow \rightarrow \text{transaction demand} \uparrow \rightarrow r \uparrow$
- **Intersection** = r aur Y ka **simultaneous** equilibrium
- **Fiscal policy** → **IS shift** | **Monetary policy** → **LM shift**
- **Slope rule**: Investment jitna interest-elastic, IS utni **flatter**. Interest-inelastic investment → **IS steeper**

7. Policy Effectiveness — Exam Ka Favourite

Situation	LM Shape	Monetary Policy	Fiscal Policy
Liquidity Trap (Keynesian range)	Horizontal	Fully INEFFECTIVE — $M_s \uparrow$ se r girta hi nahi, log sab cash hold kar lete hai (speculative demand perfectly elastic)	FULLY EFFECTIVE — poora multiplier effect, zero crowding out
Classical Range	Vertical	FULLY EFFECTIVE	100% CROWDING OUT — $G \uparrow$ se sirf r badhta, Y unchanged
Intermediate (normal case)	Upward sloping	Partially effective	Partially effective, partial crowding out
Steep IS (I interest-inelastic)	—	Weak	Strong

8. Saving – Investment Equality : Classical vs Keynes

Aadhar	Classical	Keynesian
Equilibrating variable	Rate of Interest	Level of Income
Functions	S aur I dono interest ka function	$S = f(Y)$; $I = f(MEC, r)$
Mechanism	$S > I \rightarrow r \text{ girta} \rightarrow I \uparrow, S \downarrow \rightarrow \text{equality}$	$S > I \rightarrow \text{income girta} \rightarrow S \text{ apne aap ghat kar } I \text{ ke barabar}$
Result	Equality hamesha full employment par	Equality underemployment par bhi possible

⚠ EXAM TRAP ALERT — Ye 6 Cheezein Log Sabse Zyada Galat Karte Hai

- **Original Phillips Curve** = money WAGE inflation, price inflation NAHI. Price version Samuelson-Solow (1960) ka hai
- **Post-Keynesians money supply ko ENDOGENOUS** maante hai (Classical/Monetarist = exogenous)
- **Samuelson mein path $\alpha\beta$ ki value** se tay hota hai, sirf α ya β se nahi. ($\alpha=0.5, \beta=2 \rightarrow \alpha\beta=1 \rightarrow \text{constant amplitude}$)
- **Liquidity Trap vs Classical Range** ulta yaad karna — sabse common mistake. Yaad rakho: **Trap = Fiscal** jeeta, **Classical = Monetary** jeeta
- **Ratchet Effect $\neq$ Demonstration Effect.** Ratchet = time (past peak income), Demonstration = space (padosi)
- **Hicks ka Floor** = autonomous investment + depreciation-limited disinvestment. **Ceiling** = full employment. Ulta mat karna

📖 One-Line Memory Hooks

- **Hawtrey** → Money hi sab kuch • **Kaldor** → Non-linear curves • **Hicks** → Ceiling-Floor • **Samuelson** → Free M-A • **Schumpeter** → Innovation
- **Friedman** → Permanent • **Duesenberry** → Relative • **Modigliani** → Life Cycle • **Keynes** → Absolute
- **IS = goods, LM = money** — "Investment-Saving" aur "Liquidity-Money"